

Two way ANOVA

Two-Way ANOVA

Two-way ANOVA is used to compare group means based on **two factors** and to study their interaction, for example, comparing a teacher's performance across different sections (A, B, C) and teaching methods (lecture vs activity-based).

Two-Way ANOVA Example (Sections + Teaching Method)

Section	Lecture Method	Activity-Based Method
A	72, 75	78, 80
B	80, 82	85, 87
C	76, 78	79, 81

General Procedure.

H_0 : Mean of all factor A are equal.

H_1 : Mean of all factor A are not equal.

H_0 : Mean of all factor B are equal.

H_1 : Mean of all factor B are not equal.

Test Statistics:

$$F = \frac{MS_{br}^2}{MS_w^2}$$

$$F = \frac{MS_{bc}^2}{MS_w^2}$$

Critical region: $F \geq F_{\alpha(r-1), (r-1)(c-1)}$

$F \geq F_{\alpha(c-1), (r-1)(c-1)}$

Computation:

Source of Variation	Sum of Squares (SS) Formula
Total (SST)	$\sum (X_{ij} - \bar{X}_{..})^2$
Factor A (SSA)	$b \sum (\bar{X}_{i.} - \bar{X}_{..})^2$
Factor B (SSB)	$a \sum (\bar{X}_{.j} - \bar{X}_{..})^2$
Error (SSE)	$\sum (X_{ij} - \bar{X}_{i.} - \bar{X}_{.j} + \bar{X}_{..})^2$

Or use

- 1) Take sum of all values
- 2) Take sum of Square values
- 3) Correction Factor $C.F = \frac{(\sum X)^2}{n}$
- 4) Total variation : $\sum X^2 - C.F$
- 5) Between Column variable sample : $\frac{\sum X_j^2}{r} - C.F$
- 6) Between row variable sample : $\frac{\sum X_i^2}{j} - C.F$
- 7) Within SS= Total -RowSS-ColumnSS (Step 4-Step5-Step6)
- 8) Compute F.

Source of Variation	Df	Sum of Square	Mean Square	F
Column variable	c-1	Step5	SSC/c-1	F=MSC/MSW
Row Variable	r-1	Step6	SSR/r-1	F=MSR/MSW
Within	(c-1)*(r-1)	Step7	SSw/(c-1)*(r-1)	
Total	n-1	Step 4		

Example:

Four Experimenters determine the moisture content of sample of a powder, each man taking a sample of 6 consignments their assessments are:

Experimenters	Consignments					
	1	2	3	4	5	6
1	9	10	9	10	11	11
2	12	11	9	11	10	10
3	11	10	10	12	11	10
4	12	13	11	14	12	10

Perform a two-way analysis of variance on these data. Level of significance.

General Procedure.

H_0' : Mean of all consignments are equal.

H_1' : Mean of all consignments are not equal.

H_0'' : Mean of all experimenters are equal.

H_1'' : Mean of all experimenters are not equal.

Test Statistics:

$$F = \frac{MS_{br}^2}{MS_w^2}$$

$$F = \frac{MS_{bc}^2}{MS_w^2}$$

Critical region: $F \geq F_{\alpha(r-1),(r-1)(c-1)} \rightarrow F \geq F_{0.05,(4-1),(4-1)(6-1)} \rightarrow F \geq F_{0.05,(3,15)} \rightarrow F \geq 3.29$

$F \geq F_{\alpha(c-1),(r-1)(c-1)} \rightarrow F \geq F_{\alpha(4,15)} \rightarrow F \geq 3.06$ if CR satisfied reject H_0

Experimenters	Consignments					
	1	2	3	4	5	6
1	9	10	9	10	11	11
2	12	11	9	11	10	10
3	11	10	10	12	11	10
4	12	13	11	14	12	10

Computation:

- 1) Take sum of all values
- 2) Take sum of Square values

Experimenters	Consignments						Row sum
	1	2	3	4	5	6	
1	9	10	9	10	11	11	60
2	12	11	9	11	10	10	63
3	11	10	10	12	11	10	64
4	12	13	11	14	12	10	72
Column sum	44	44	39	47	44	41	259
Square sum	490	490	383	561	486	421	2831

3) Correction Factor C.F = $\frac{(\sum X)^2}{n} = \frac{(44+44+39+47+44+41)^2}{24} = 2795.04167$

4) Total variation : $\sum X^2 - C.F = 2831 - 2795.04167 = 35.95833$

5) Between Column variable sample : $\frac{\sum X_j^2}{r} - C.F = \left(\frac{44^2}{4} + \frac{44^2}{4} + \frac{39^2}{4} + \frac{47^2}{4} + \frac{44^2}{4} + \frac{41^2}{4} \right) - 2795.04167 = 9.708333$

6) Between row variable sample : $\frac{\sum X_i^2}{j} - C.F = \left(\frac{60^2}{6} + \frac{63^2}{6} + \frac{64^2}{6} + \frac{72^2}{6} \right) - 2795.04167 = 13.1250$

7) Within SS = Total - RowSS - ColumnSS (Step 4 - Step5 - Step6) = $35.95833 - 9.708333 - 13.1250 = 13.1250$

8) Compute F.

Source of Variation	Df	Sum of Square	Mean Square	F
Consignment	c-1=5	9.7083	$9.7083/5=1.94166$	$F=MSC/MSW=1.94166/0.875$ 2.21904
Experimenter	r-1=3	13.1250	$13.125/3=4.375$	$F=MSR/MSW=4.375/0.875=5$
Within	$(c-1)*(r-1)=15$	13.1250	$13.125/15=0.875$	
Total	n-1=24-1=23	35.9583		

$2.21904 > 2.90 \rightarrow$ doesn't satisfied

$5 > 3.29 \rightarrow$ satisfied

Conclusion: 1) As Critical region doesn't satisfied therefore there is no evidence to reject H_0 at 0.05 level of significance we conclude that all consignment have same average, As Cr satisfied therefore there is evidence to reject H_0 at 0.05 level of significance we conclude that all experimenter have different average response.